Logical Knowledge in KGs

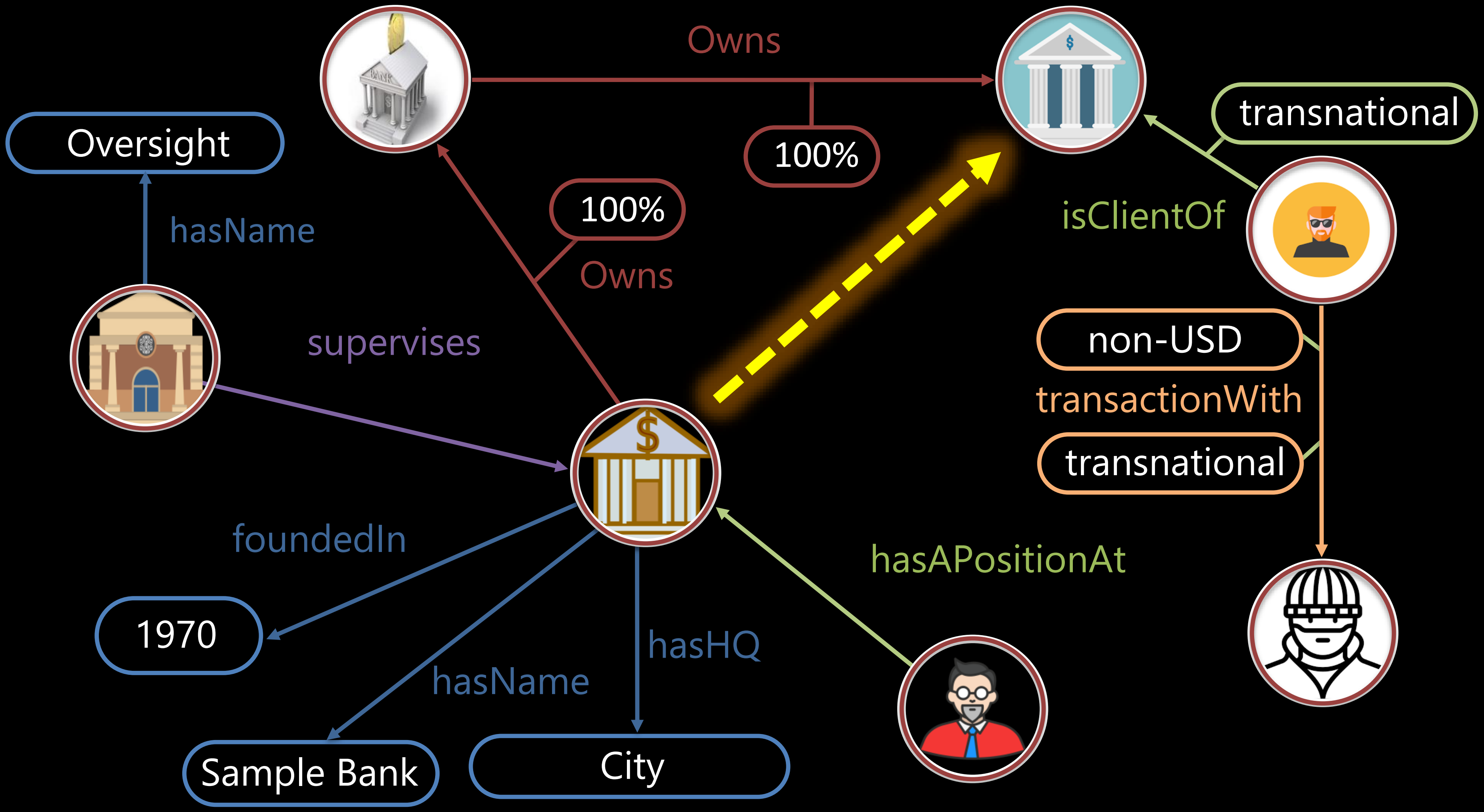
Challenges: Recursion and Creation

Emanuel Sallinger

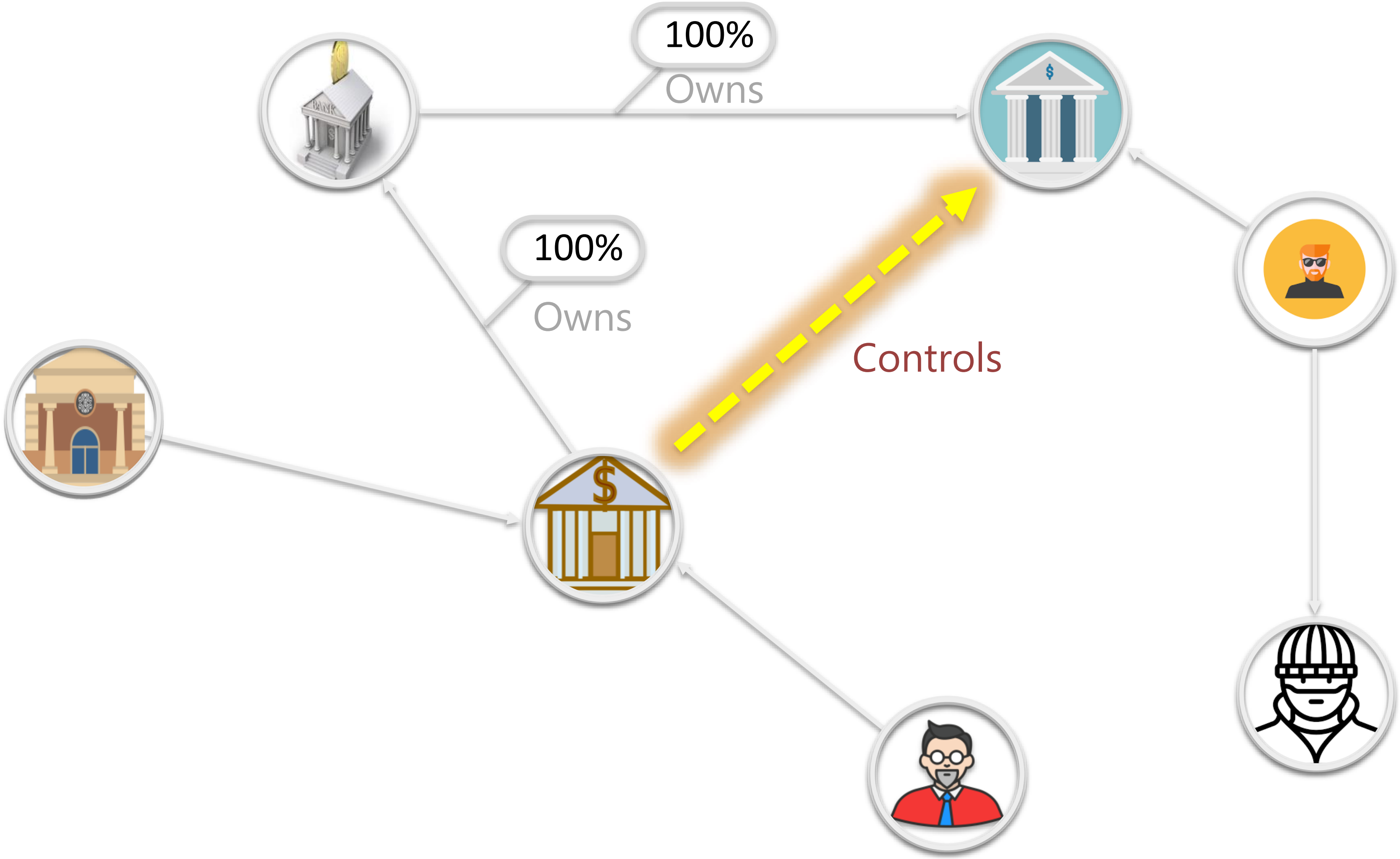
Main Challenges

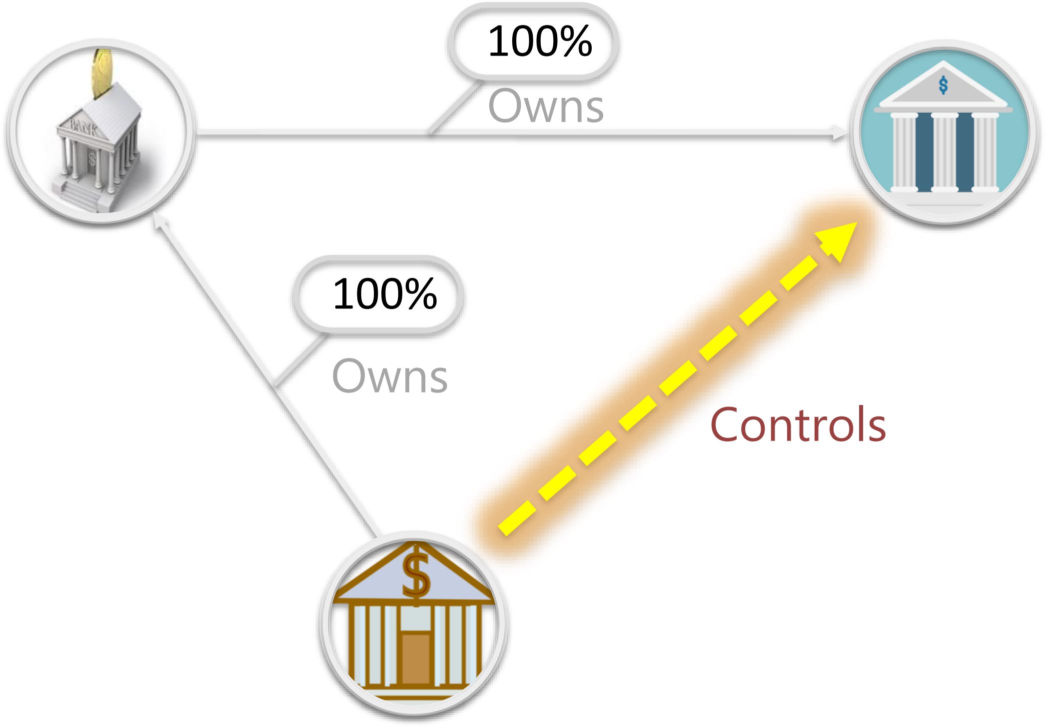
1. **Recursion**
   unlimited graph exploration

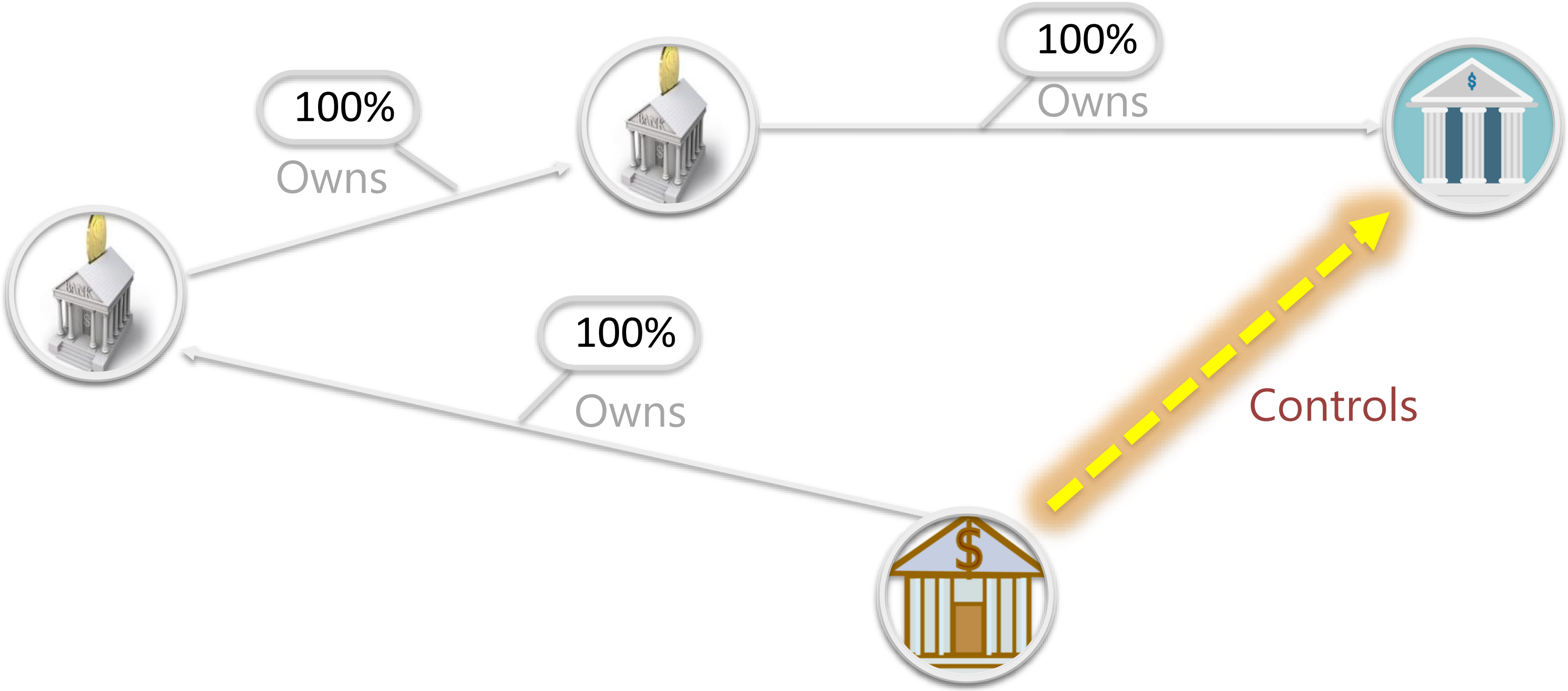
2. **Object Creation**
   exploring unknown parts of the KG

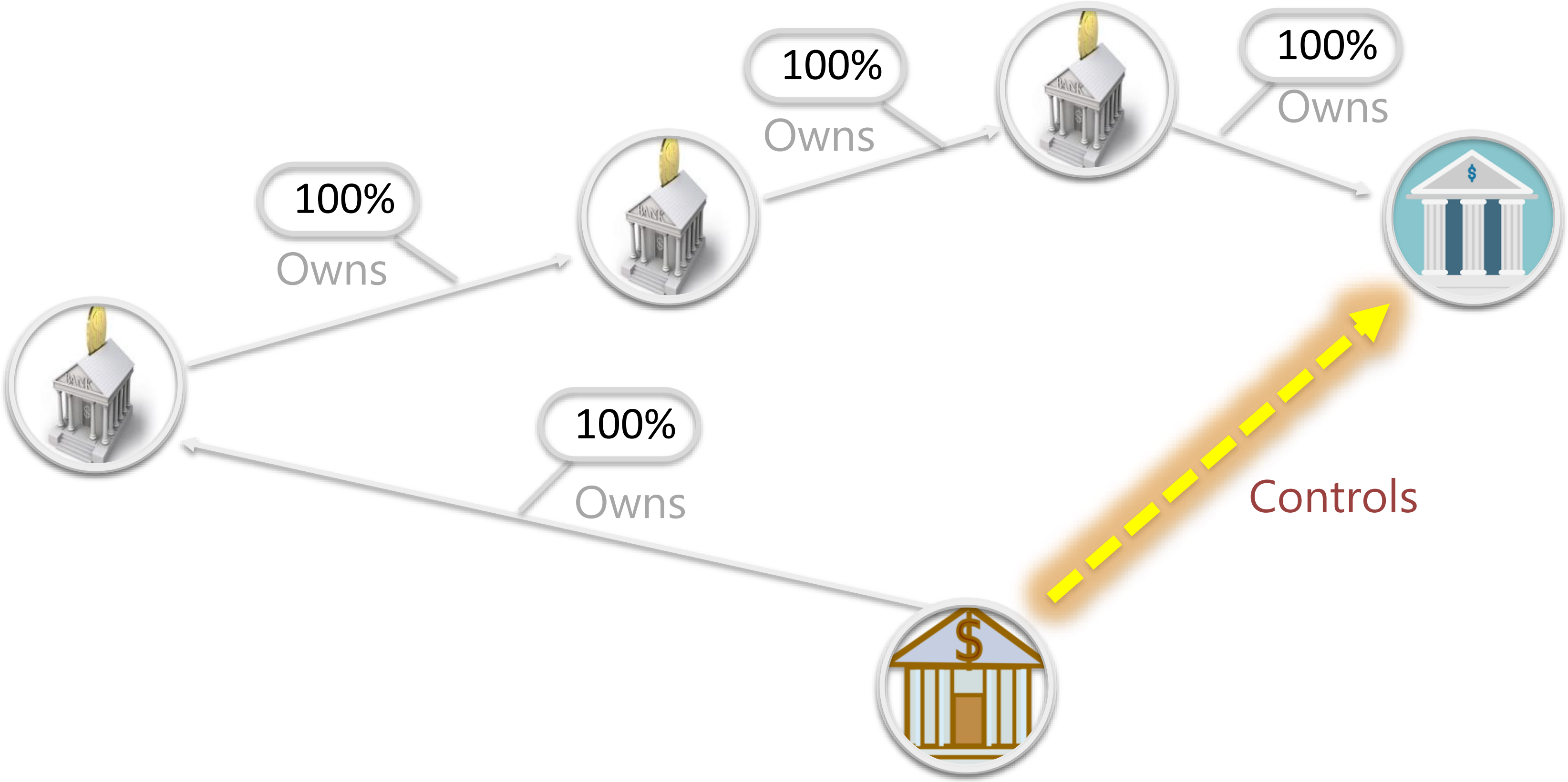


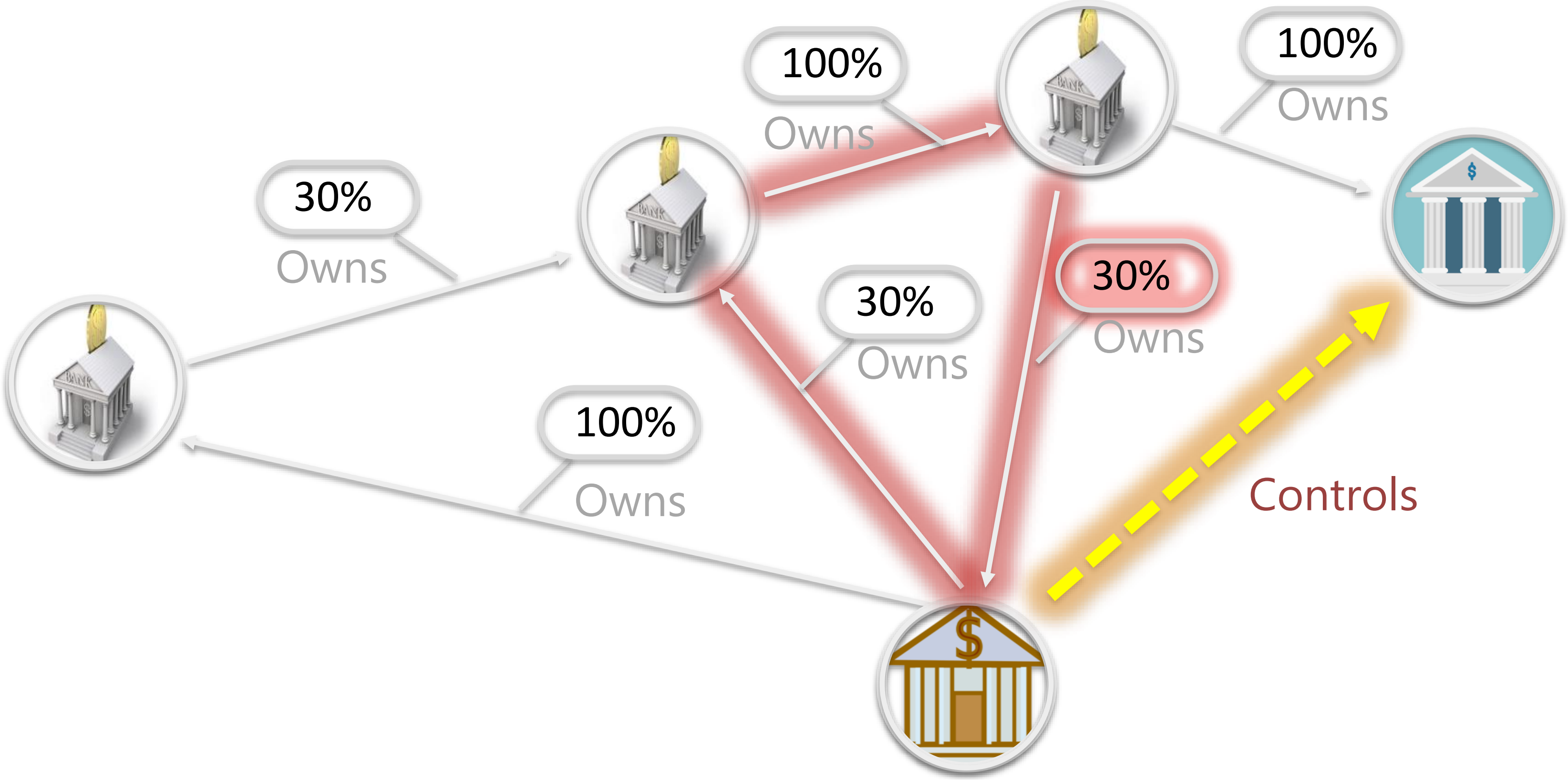
Owns
100%
transnational
Oversight
100%
hasName
isClientOf
Owns
supervises
non-USD
transactionWith
transnational
foundedIn
hasAPositionAt
1970
hasHQ
hasName
Sample Bank
City


100%
Owns
100%
Owns
Controls


100%
Owns
100%
Owns
Controls


100%
Owns
100%
Owns
100%
Owns
Controls


100%
Owns
100%
Owns
100%
Owns
100%
Owns
Controls

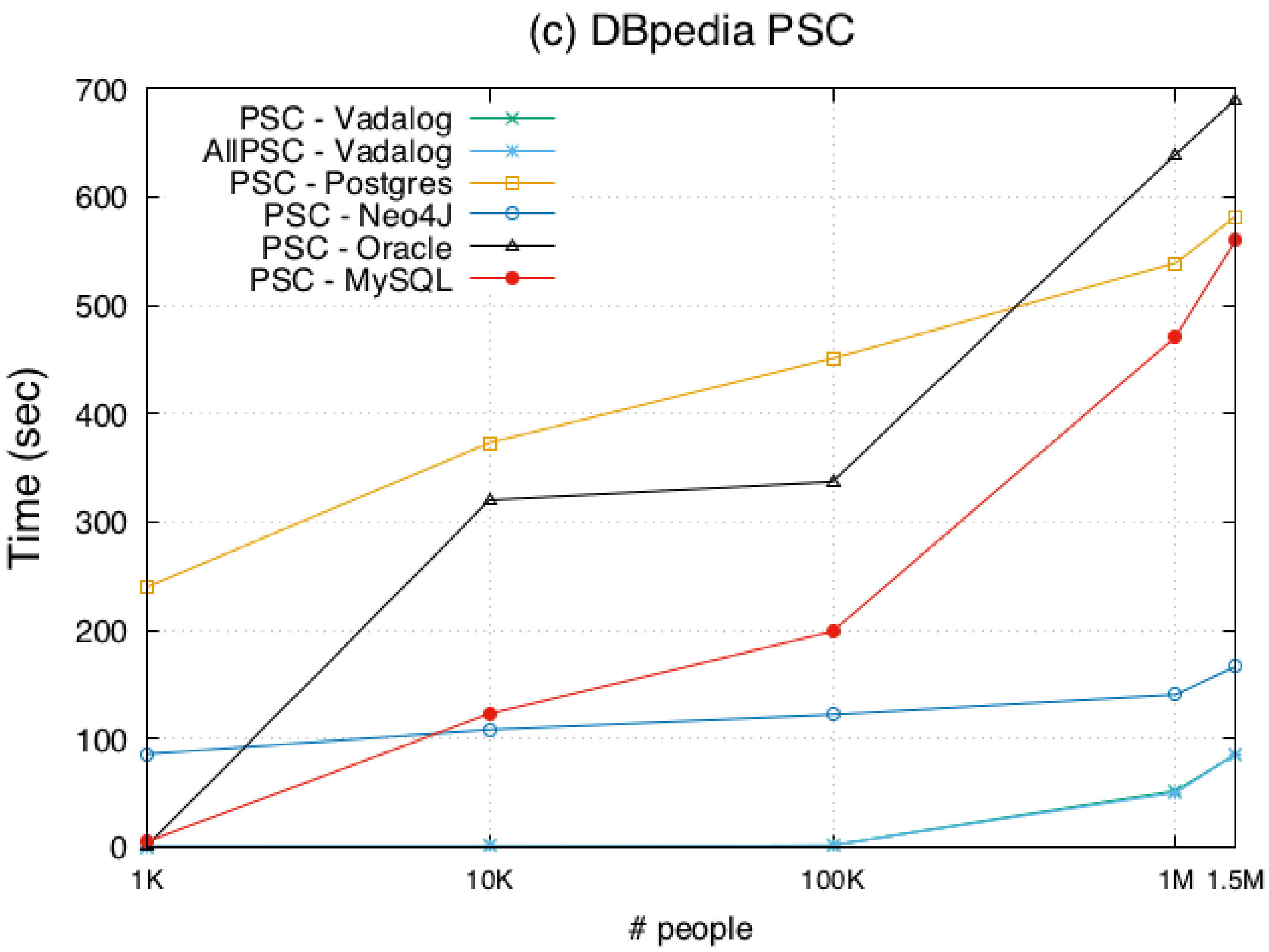

100%
Owns
100%
Owns
30%
Owns
30%
Owns
30%
Owns
100%
Owns
Controls

x **controls** y if

  x **directly holds** over 50% of y, or

  x **controls a set** S of companies that **jointly control** y

```
control(X,Y) :- own(X,Y,W), W > 0.5.
control(X,Z) :- control(X,Y), own(Y,Z,W),
                V=msum(W,<Y>), V > 0.5.
```

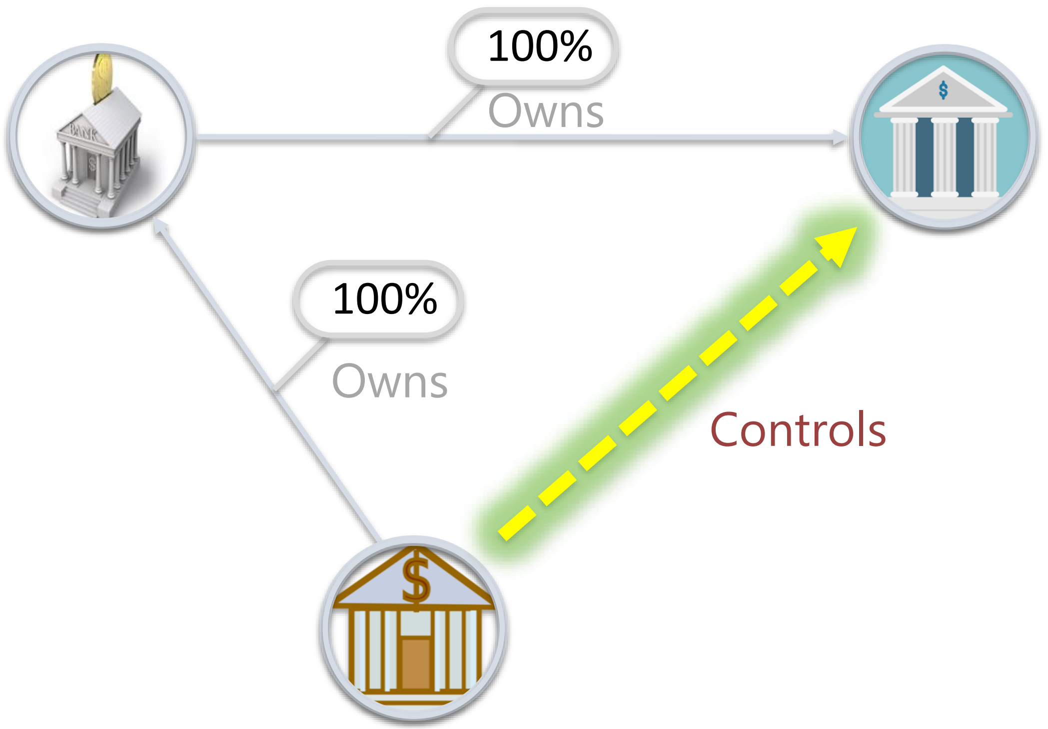


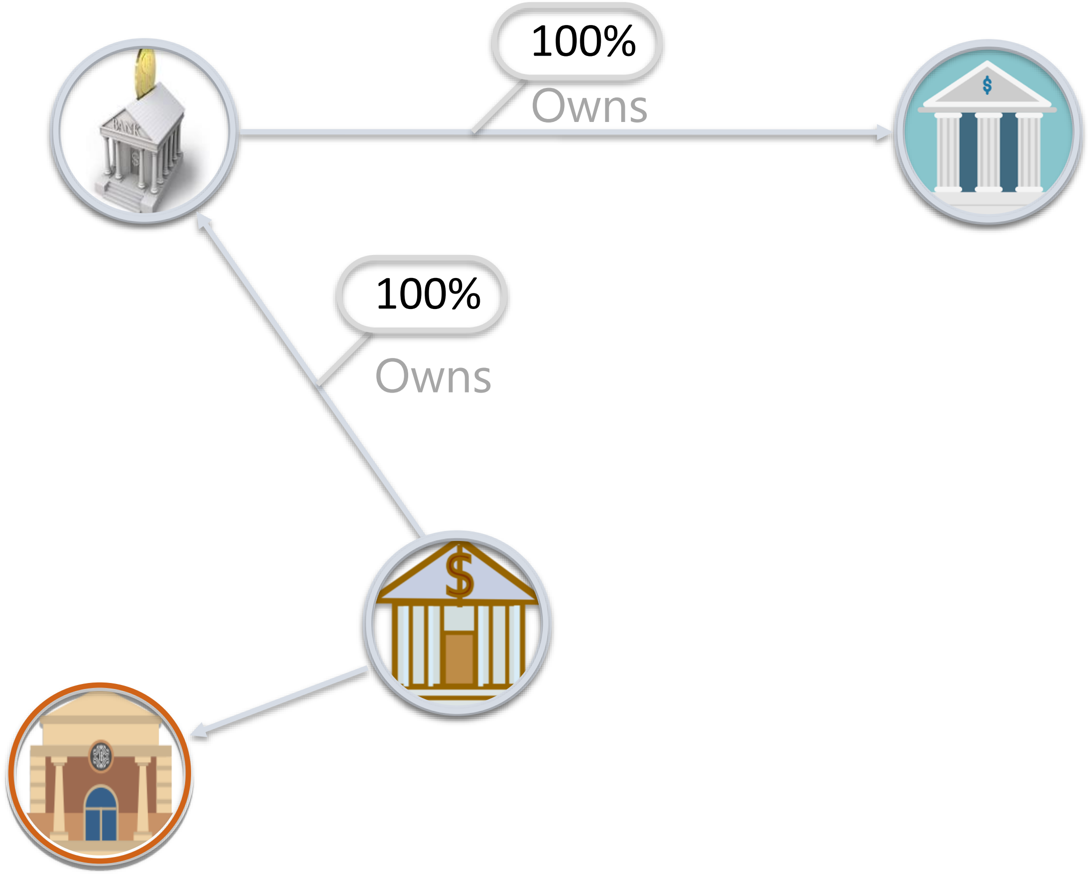
(c) DBpedia PSC
PSC - Vadalog
AllPSC - Vadalog
PSC - Postgres
PSC - Neo4J
PSC - Oracle
PSC - MySQL
Time (sec)
0
100
200
300
400
500
600
700
1K
10K
100K
1M
1.5M
people

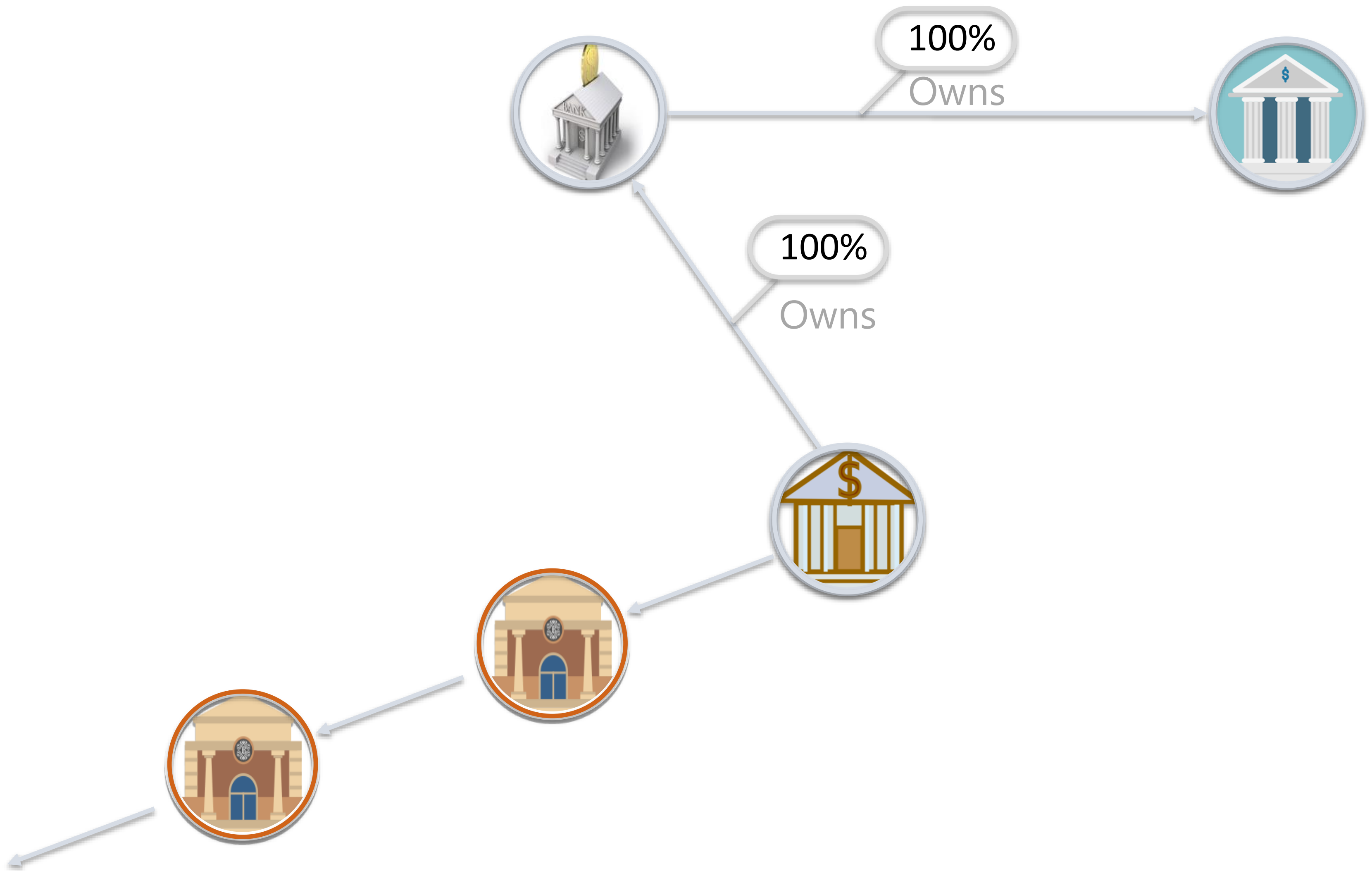
Main Challenges

1. **Recursion**
   unlimited graph exploration

2. **Object Creation**
   exploring unknown parts of the KG


100%
Owns
100%
Owns
Controls


100%
Owns
100%
Owns


100%
Owns
100%
Owns

x **controls** y if

  x **directly holds** over 50% of y, or

  x **controls a set** S of companies that **jointly control** y

```
control(X,Y) :- own(X,Y,W), W > 0.5.
control(X,Z) :- control(X,Y), own(Y,Z,W),
                V=msum(W,<Y>), V > 0.5.
```

Main Challenges

1. **Recursion**
   unlimited graph exploration

2. **Object Creation**
   exploring unknown parts of the KG

Query answering under
recursive Datalog with object creation
(existential quantification) is **undecidable**.

Logical Knowledge in KGs

Challenges: Recursion and Creation

Emanuel Sallinger